Vikram Raj Nagoor Kani

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Education

University of Illinois Urbana Champaign — Urbana, IL
Master of Engineering in Autonomy and Robotics — *Aug 2024 - present*

BS Abdur Rahman Crescent Institute of Technology — Chennai, India
Bachelor of Technology in Electronics and Communication Engineering — *Jul 2017 - Jun 2021*

Work Experience

Robotics Systems Integration Engineer Intern, *UIUC College of Veterinary Medicine* — *Aug 2025 - present*

- Designing a **ROS 2** based automation pipeline hosted in **Docker Container** for grain quality inspection, integrating multi-sensor systems with existing grain probe infrastructure **improving speed, cost, accuracy, hours of operation and safety** compared to manual operation of the probe.

Robotics Machine Learning Engineer Apprentice, *Kohler* — *Jun 2025 - Aug 2025*

- Developed a package validation system using a **lightweight instance segmentation model**, capable of **verifying** product presence every **30 ms on an edge device**, with seamless integration into the defect inspection pipeline.

Machine Learning Engineer Intern, *UIUC College of Veterinary Medicine* — *Jun 2025 - Aug 2025*

- Analyzed and trained **SOTA pose detection models** on homogeneous vs. heterogeneous environment swine datasets to evaluate their effectiveness at keypoint detection in **real-world** farm settings, contributing to an upcoming **research publication**.

Robotics Machine Learning Engineer Intern, *UIUC College of Applied Health Sciences* — *Jan 2025 - May 2025*

- Optimized **multimodal deep learning** pipeline for **STRETCH Robot AI**, enabling compatibility with **low-end GPUs** by reducing VRAM usage from 12GB to 7GB .
- Improved the reliability and safety of elderly-assistive object pick-and-place tasks, **doubling** the pickup **success** rate from **20% to 40%**.

Automation Engineer, *Accenture* — *Aug 2021 - Jul 2024*

- Implemented **CI/CD pipelines** for provisioning and managing **Docker Containers**, **Azure Cloud** and **BareMetal servers**, **reducing manual** setup and maintenance effort by up to **60%**.
- Evaluated **cognitive vision models** on **IoT edge** servers to assess their accuracy in detecting and reporting faulty surveillance cameras at Microsoft data centers.

Embedded System Engineer Intern, *Nokia* — *Feb 2021 - May 2021*

- Developed an **IoT device** to **detect obstacles** blocking accessibility of fire extinguishers and alert security in **real time** for Nokia's Manufacturing Factory as part of their safety measures.

Projects

F1Tenth Robust Navigation using LiDAR and Camera

ROS, Python, Open CV, NVIDIA Jetson, F1 Tenth, Unity, Linux

- Introduced a **light weight** lane detection algorithm using **OpenCV**, optimizing **computational speed by 30%** on **Nvidia Jetson** for **real-time waypoint generation** and accepted as **open source contribution** for UIUC ECE 484.
- Developed a **photorealistic digital twin** of the **F1TENTH** car and environment using a Unity-based open-source simulator, enabling testing and validation with **minimal sim-to-real gap** and accelerating development.

6-DOF Robot Arm Pose Prediction Using Deep Learning

PyTorch, Python, OpenCV, ResNet, YOLO

- Developed a **deep learning model** to predict **6 Degrees of Freedom** for **robotic arms** to pick up warehouse parts. Using a combination of RGB and Depth images in a **dual-stream U-Net** architecture with hybrid **feature fusion**.
- Developed a **custom data loader** and image augmentation pipeline to convert BOP dataset format to YOLO format for seamless integration with YOLO-based models.

Persistent Pedestrian Detection Using Sensor Fusion

GEM E4, PyTorch, ROS, Python, Open CV, YOLO, GEMStack

- Developed **persistent pedestrian detection algorithm** utilizing camera and LiDAR data from GEM e4 vehicle optimized for **real-time systems** by implementing techniques like **voxel down-sampling** reducing time complexity.

Digital Image-to-Drawing Replication Using UR3e Robot Arm

UR3, ROS, Python, Open CV

- Developed an **OpenCV-based pipeline** to detect and refine image contours, **preserving key features** while **reducing waypoints** to improve robotic drawing speed.

Technical Skills

Programming Languages : Python, C++, Bash, PowerShell, LabView, Matlab, Azure CLI.
Technologies/Frameworks : Pytorch, Open CV, ROS, ROS2, Docker, Roboflow, Anaconda , NVIDIA Jetson, GEM E2, GEM E4, F1Tenth, Azure, Gazebo, Gazebo Ignition, Unity, Git, Linux, YOLO, MASKRCNN, HRNet, Dino, Detic, SigLIP, Open AI Gym, Deep Learning, Machine Learning, UR3e, CI/CD, SAM, Transformers, ResNet.
Coursework : Principles of Safe Autonomy, Deep Learning with Computer Vision, Autonomous Vehicle System Engineering.